

Duck Agents

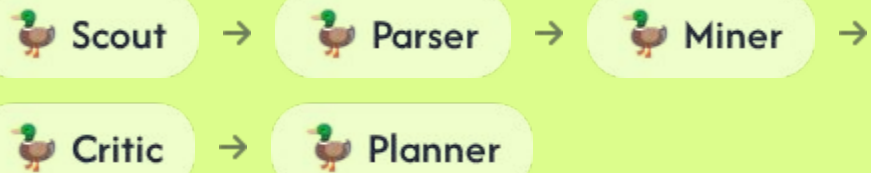
Team Duck Agent: 42xNeedle Hackathon 2026

Meet the Duck Agent Flock

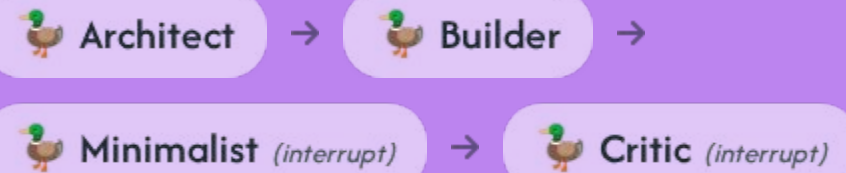
 Scout Parser Miner Planner Minimalist Architect Builder Tactician Tester Debugger Repairer Critic Validator Logger Coordinator Guardrail

Collaboration Protocol

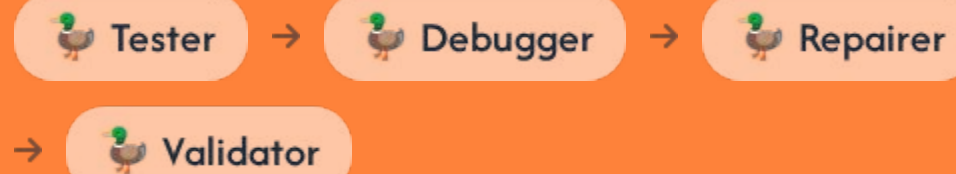
1 Phase 1 — Understand



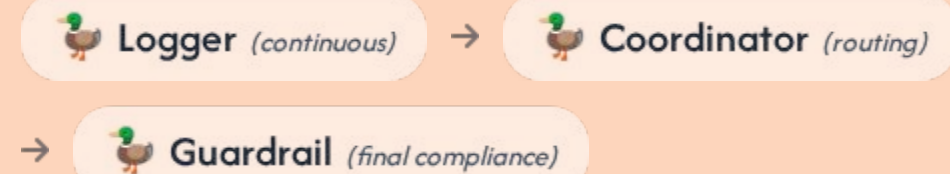
2 Phase 2 — Build



3 Phase 3 — Verify



4 Phase 4 — Govern



Decoding the Orchestrator Log

The Mastermind

Phase Switching: The Orchestrator acts as the brain, seamlessly transitioning between stages like *Understand* and *Verify*.

Role Assignment: Instead of a single AI trying to do everything, it summons specialized *duck* agents for targeted tasks.

The Goal: Write a perfect C *itoa* function entirely autonomously!



Phase 1: Understand



Scout & Parser

These agents scan the environment and parse the prompt to understand the ultimate goal: creating an *itoa* function in C.



Miner

Digs deep into the parsed data to extract the core constraints and logic: handling specific bases, pointers, and negative numbers.



Coordinator

Manages the flock, ensuring all prerequisites and rules are met before handing off the blueprint to the execution phase.

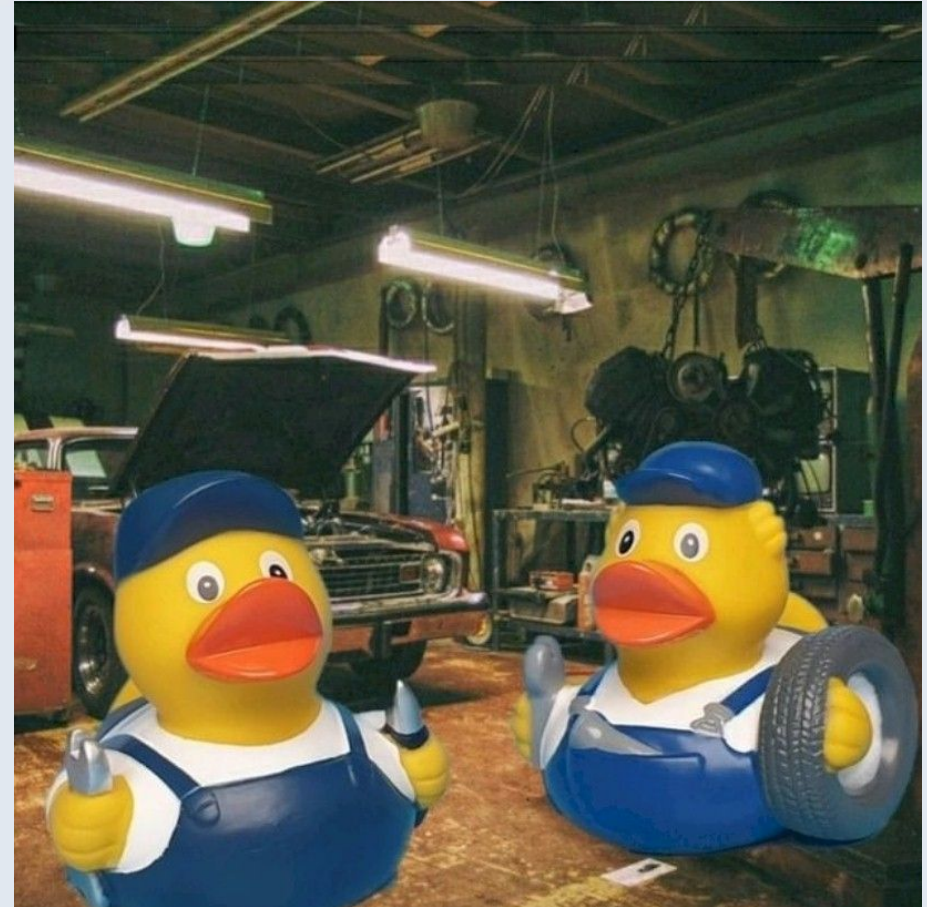
Phase 2: Execution & Code Generation

Writing Code

Once the 'Understand' phase finishes, agents trigger the **WRITE_FILE** action to generate *itoa.c*.

“But wait! The very first draft lacks a main() function, making it impossible to compile and test.”

The Orchestrator immediately notices the gap in logic and shifts gears.



Phase 3: Verify & Heal



Tester

Takes the generated *itoa.c*, adds a `main()` function, and compiles it using the `RUN_TESTS` action to see if it executes.



Debugger

Catches the errors! It analyzes the test output and discovers undefined behavior because a crucial return statement was missing.



Repairer

Rewrites the file iteratively based on the debug logs. It's an autonomous, self-healing loop striving for perfection!

Questions?

Team Duck Agent